

How the Web Works

Learning Objectives

- Explain the client-server model and how web pages are delivered
- Describe the role of IP addresses, DNS, and HTTP in web communication
- Differentiate between static and dynamic web pages

Engage (5-7 minutes)

When you type a website address, something remarkable happens in milliseconds—the page appears. But how does your browser know where to find it? Today we uncover the invisible infrastructure that makes the web function.

Explore (10-12 minutes)

Trace the journey of loading a webpage: your browser contacts a DNS server to translate “example.com” into an IP address, sends an HTTP request to the web server, receives HTML/CSS/JavaScript files, and renders the page. Draw this flow as a diagram with labeled steps.

Explain (10-12 minutes)

The client-server model is the foundation of the web. Your browser (client) requests resources from web servers using HTTP/HTTPS protocols. DNS translates human-readable domain names into numerical IP addresses. Static pages are fixed HTML files served as-is; dynamic pages are generated on-the-fly using server-side languages like Python or PHP.

Elaborate (8-10 minutes)

A small business wants a website. Compare static hosting (HTML files only) versus dynamic hosting (with database and server-side code). Explain when each option is appropriate and what technologies each requires.

Evaluate (5-8 minutes)

Quiz: Label the steps in the client-server request cycle, define IP address and DNS, and explain one difference between static and dynamic web pages with an example of each.